AARYAN PATEL

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EDUCATION

University of California San Diego — San Diego, CA

Bachelor of Science in Artificial Intelligence (GPA: 4.0) — *Expected Graduation June 2029*

- **Relevant Coursework**: CSE 29 (Operating Systems, Memory Management), CSE 30 (Computer Architecture, Assembly Language), Data Structures & Algorithms (Java), Advanced Data Structures & Algorithms (**C++**)

EXPERIENCE

Software Developer — January 2026 – Present

Beernerd — *La Jolla, CA*

- Architected a scalable backend on **AWS EC2**, utilizing **Nginx** and **Route 53** to host a secure **Fastify (Node.js) REST API** with **99+% uptime** interfacing with **Supabase (SQL)** and integrating **Gemini AI**.
- Led a **database migration** to a polymorphic **SQL** schema, resolving foreign key constraints to enable the platform to scale from one game to multiple activity types.
- Labeled gameplay footage and fine-tuned a custom **YOLOv8** model for **real-time object detection**, laying groundwork for automated scoring.

CTO of Symmyo — October 2025 – Present

Blackstone Launchpad — *La Jolla, CA*

- Won **1st place** in Healthcare Track for developing a **real-time EMG signal processing** prototype that captured, filtered, and streamed muscle activation data in under 24 hours.
- Engineered a **Python preprocessing pipeline** (bandpass, rectification, low-pass filters) that reduced EMG noise by **80%**, enabling **real-time visualization** and live signal analysis.
- Trained a **Random Forest** classifier on the SEMG DB1 dataset and evaluated it with **Leave-One-Subject-Out cross-validation** to measure generalization to unseen users.

Computer Vision Researcher — June 2024 – November 2024

Inspirit AI — *Oroville, CA*

- Created custom **CNN** architectures and focal loss function for **real-time emotion detection**, achieving **92.05%** accuracy on color images and **83.11%** on grayscale.
- Analyzed emotion trends across movie content and found higher-rated titles (4–5 stars) coincided with **sadness dropping from 20% to 12%**; added batch norm and dropout to raise validation accuracy **12%**.

PROJECTS

ROS2-Based Autonomous Drone (Vertical Flight Society) | *ROS2, **C++**, OpenCV, PX4* — January 2026 – June 2026

- Engineered a **real-time computer vision pipeline** using **ROS2** and **C++** with **OpenCV/YOLOv8** to identify and classify cargo with **90+%** detection accuracy.
- Implemented a **state machine** synchronizing vision data with flight controls, enabling autonomous **robotics** transport of 5 sandbags at a time to a designated drop zone.

CRED Website Revamp, Backend Lead | *Next.js, Express.js, MongoDB, React, TypeScript* — December 2025 – May 2026

- Led the **backend** of a production-grade website revamp for a team of 4 within Triton Software Engineering, working across **Express.js** and **MongoDB**.
- Designed strict **Mongoose schemas** with **TypeScript** for end-to-end type safety and integrity of sensitive fields such as CDCR numbers and demographics.

Corro (Swans Applied AI Hackathon, 2nd Place, $1,500) | *Python, FastAPI, SQLite, OpenAI API* — October 2026

- As **lead developer** on a team of 2, built a **FastAPI** and **SQLite** app that imports a Clio matter read-only (GET-only, enforced in code) into a searchable knowledge graph of **234 items** and **2,953 statements**.
- Verified **2,781 quotes** verbatim against source pages in code, with search in **under 1 ms**.

TECHNICAL SKILLS

Languages: **Python**, **SQL**, **JavaScript**, TypeScript, **HTML/CSS**, C++, C, Java, Assembly
Frameworks: Express.js, FastAPI, React, Next.js, ROS2, Flask, Django, PyTorch, TensorFlow
Cloud & Tools: **Raspberry Pi**, **Docker**, Git, Linux, AWS EC2, Nginx, Route 53, Google Cloud Platform, VS Code
Libraries & ML: Pandas, NumPy, OpenCV, Scikit-Learn, Matplotlib, RAG, Prompt Engineering